Example 6.1

For the measures in the figure, compute sine, cosine and tangent ratios of the angle θ.

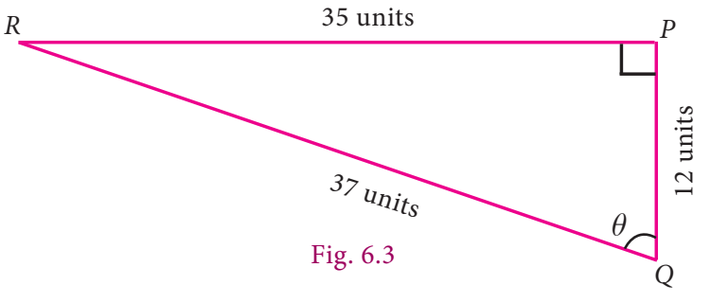


Fig. 6.3

Solution

In the given right angled triangle, note that for the given angle θ, PR is the 'opposite' side and PQ is the 'adjacent' side.

$$\sin\theta = \frac{opposite\,side}{hypotenuse} = \frac{PR}{QR} = \frac{35}{37}$$

$$\cos\theta = \frac{adjacent\,side}{hypotenuse} = \frac{PQ}{QR} = \frac{12}{37}$$

$$\tan\theta = \frac{opposite\,side}{adjacent\,side} = \frac{PR}{PQ} = \frac{35}{12}$$

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It is enough to leave the ratios as fractions. In case, if you want to simplify each ratio neatly in a terminating decimal form, you may opt for it, but that is not obligatory.

Note

- Since trigonometric ratios are defined in terms of ratios of sides, they are unitless numbers.
- Ratios like $\sin\theta$, $\cos\theta$, $\tan\theta$ are not to be treated like $(\sin)\times(\theta)$, $(\cos)\times(\theta)$,$(\tan)\times(\theta)$.

Thinking Corner

The given triangles ABC,DEF and GHI have measures 3-4-5, 6-8-10 and 12-16-20.

Are they all right triangles?

How do you know?

The angles at the vertices B, E and H are of equal size (each angle is equal to θ).

With these available details, fill up the following table and comment on the ratios that you get.

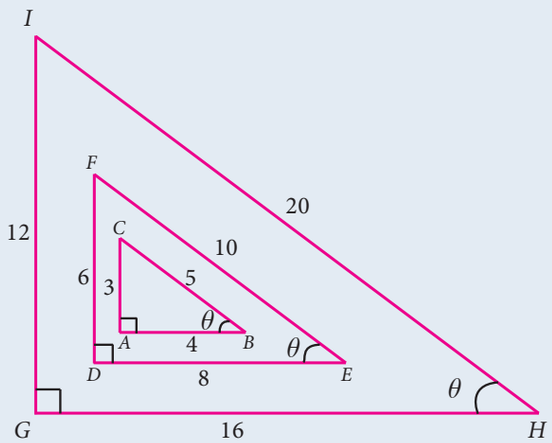


Fig. 6.4

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